Brain, Behavior, and Immunity 118 (2024) 192–201

Contents lists available at ScienceDirect

Brain Behavior and Immunity

journal homepage: www.elsevier.com/locate/ybrbi

Full-length Article

Omega-3 fatty acids for inflamed depression – A match/mismatch study

Klara Suneson [a,b], Gustav Söderberg Veibäck [a,c], Jesper Lindahl [a,d], Johanna Tjernberg [a,e], Darya Ståhl [a], Simon Ventorp [a], Filip Ängeby [a,d], Karl Lundblad [d,f], Owen M. Wolkowitz [g], Daniel Lindqvist [a,e,*]

[a] Unit for Biological and Precision Psychiatry, Department of Clinical Sciences Lund, Lund University
[b] Department of Adult Psychiatry, Office for Psychiatry, Habilitation and Technical Aids, Malmö, Sweden
[c] Department of Gastroenterology and Nutrition, Department of Clinical Sciences, Skåne University Hospital, Malmö, Sweden
[d] Department of Adult Psychiatry, Office for Psychiatry, Habilitation and Technical Aids, Lund, Sweden
[e] Psychiatry Research Skåne, Office for Psychiatry, Habilitation and Technical Aids, Lund, Sweden
[f] Office for Psychiatry, Norra Stockholm Psykiatri, Region Stockholm, Sweden
[g] Department of Psychiatry and Behavioral Sciences and Weill Institute for Neurosciences, University of California, San Francisco, CA, USA

ARTICLE INFO

Keywords:
Major depressive disorder
N-3 PUFAs
Inflammation

ABSTRACT

Despite decades of research on the pathophysiology of depression, the development of new therapeutic interventions has been slow, and no biomarkers of treatment response have been clinically implemented. Several lines of evidence suggest that the clinical and biological heterogeneity among patients with major depressive disorder (MDD) has hampered progress in this field. MDD with low-grade inflammation – "inflamed depression" – is a subtype of depression that may be associated with a superior antidepressant treatment response to anti-inflammatory compounds. Omega-3 fatty acid eicosapentaenoic acid (EPA) has anti-inflammatory properties, and preliminary data suggest that it may be particularly efficacious in inflamed depression. In this study we tested the hypothesis that add-on EPA has greater antidepressant efficacy in MDD patients with high baseline high-sensitivity C-reactive protein (hs-CRP) compared to MDD patients with low hs-CRP. All subjects received 2.2 g EPA, 400 mg docosahexaenoic acid and 800 mg of other fatty acids daily for 8 weeks, added to stable ongoing antidepressant treatment. The primary outcome was change in the 17-item Hamilton Depression Rating Scale (HAMD-17). Patients and raters were blind to baseline hs-CRP status. In an intention-to-treat analysis including all subjects with at least one post baseline visit (n = 101), a hs-CRP cut-off of ≥1 mg/L, but not ≥3 mg/L, was associated with a greater improvement in HAMD-17 total score. In addition to a general antidepressant effect among patients with hs-CRP ≥ 1 mg/L, adjuvant EPA treatment improved symptoms putatively related to inflamed depression such as fatigue and sleep difficulties. This adds to the mounting evidence that delineation of MDD subgroups based on inflammation may be clinically relevant to predict treatment response to anti-inflammatory interventions.

1. Introduction

Omega-3 polyunsaturated fatty acids (n-3 PUFAs) have shown an antidepressant effect in some, but not all, studies in major depressive disorder (MDD) (Mocking et al., 2016; Appleton et al., 2021). Effect sizes have so far been too small for broad clinical introduction, but several factors have been identified as predictors of a greater clinical response. Such factors include add-on treatment (as opposed to monotherapy) and high content of n-3 PUFA eicosapentaenoic acid (EPA) relative to docosahexaenoic acid (DHA) (Mocking et al., 2016; Liao et al., 2019). While the mechanisms underlying a putative antidepressant effect of n-3 PUFAs are not fully understood, they may involve an interplay between anti-inflammatory and neuroprotective factors (Borsini et al., 2021). Indeed, n-3 PUFAs have various anti-inflammatory properties resulting in several downstream effects such as decreased production of inflammatory signaling molecules eicosanoids (e.g. prostaglandins) (Calder, 2017) and increased synthesis of pro-resolving lipid mediators resolvins and protectins (Calder, 2013). Given these mechanisms of action, n-3 PUFAs may have a more robust antidepressant effect in patients with signs of inflammation compared to non-inflamed depression (Suneson

* Corresponding author at: Baravägen 1, SE-22185 Lund, Sweden.
E-mail address: Daniel.lindqvist@med.lu.se (D. Lindqvist).

https://doi.org/10.1016/j.bbi.2024.02.029
Received 30 September 2023; Received in revised form 31 January 2024; Accepted 27 February 2024
Available online 1 March 2024
0889-1591/© 2024 The Author(s). Published by Elsevier Inc. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

et al., 2021). Immune activation has been suggested as an important pathophysiological mechanism of MDD in some, but not all, cases (Raison and Miller, 2011; Haroon et al., 2020; Lynall et al., 2020). Although there is currently no consensus how to best define "inflamed depression", systemic low-grade inflammation, per high-sensitivity C-reactive protein (hs-CRP) cut-offs of >3 mg/L or >1 mg/L has been reported in between 30 and 60 % of depression cases (Osimo et al., 2019). Concurrent low-grade inflammation in MDD has been associated with certain clinical features such as specific depressive symptoms (e.g. anhedonia, fatigue and sleep problems), obesity and poorer response to conventional antidepressants (Miller et al., 2017; Frank et al., 2021; Suneson et al., 2023; Wijaya et al., 2022).

Data from two Randomized Controlled Trials (RCTs) suggest that monotherapy with n-3 PUFAs may be more efficacious in inflamed depression than in non-inflamed depression (Rapaport et al., 2016; Mischoulon et al., 2022). Rapaport et al. showed that EPA had a large antidepressant effect, but only in depressed subjects with signs of pre-treatment low-grade inflammation (Rapaport et al., 2016). Elevated hs-CRP was one of the strongest individual predictors of antidepressant response in this study (Rapaport et al., 2016). Mischoulon et al. evaluated the effect of different doses of EPA vs placebo in an MDD sample enriched for patients with low-grade inflammation (per increased BMI and hs-CRP) (Mischoulon et al., 2022). They reported dose-dependent medium-to-large effect sizes for clinical response which favored the EPA groups over placebo, especially at higher doses (Mischoulon et al., 2022). After 12 weeks of treatment, hs-CRP levels had decreased in the EPA groups in a dose-dependent manner compared to the placebo group, suggesting that EPA can target low-grade inflammation in depression. These two studies, along with other findings (Su et al., 2014), suggest that n-3 PUFAs with high EPA content might be more efficacious in depressed patients with low-grade inflammation compared to those without inflammation. Therefore, several investigators have proposed that new studies testing the antidepressant effect of n-3 PUFAs should use a personalized medicine approach targeting patients with increased inflammation and associated metabolic alterations, such as obesity (Su, 2015; Guu et al., 2019; Wu et al., 2023).

In the present study, we investigated if the antidepressant effect of add-on n-3 PUFAs was greater in inflamed versus non-inflamed depression. We hypothesized that patients with elevated hs-CRP levels at baseline ("inflammation group") would have a superior antidepressant response than depressed patients with low pre-treatment hs-CRP levels ("non-inflammation group").

2. Methods

2.1. Overview

Patients were recruited between 2017 and 2023 via social media ads, newspapers and radio, as well as clinical referrals. Study visits were done at the psychiatric clinic in Lund, Sweden. The study was approved by the Ethical Review Board in Lund, Sweden (ref #2017/150), and monitored according to GCP guidelines by Clinical Studies Sweden, Forum South. The study was pre-registered at clinicaltrials.gov (ref #NCT03143075).

2.2. Study design and participants

Full inclusion and exclusion criteria, and study procedure details, can be found in the study protocol which has been previously published (Suneson et al., 2022). In summary, all subjects had a current MDD episode according to the DSM-5 (APA, 2013) criteria with a symptom duration >4 weeks. At enrollment, all subjects were on a stable antidepressant treatment (duration >6 weeks), had a Hamilton Depression Rating Scale 17-items (HAMD-17) (Hamilton, 1960) score of ≥15, a Clinical Global Impression (Busner and Targum, 2007) severity score ≥3, and agreed not to significantly modify their diet during the course of the study. Bipolar or psychotic disorder were exclusion criteria, but mood stabilizers or neuroleptics were allowed in combination with antidepressants. Other examples of exclusion criteria were any current infection or other medical illness that could interfere with study results (such as autoimmune disorders, malignancy or insulin-dependent diabetes mellitus), regular medication that could confound treatment results (e.g. anti-inflammatory drugs or immunosuppressants), intake of n-3 PUFA supplements for ≥3 consecutive days in the month preceding screening, and recent initiation of psychotherapy (4 weeks before baseline) or planned changes in treatment (psychotherapy or antidepressants) during the study. Subjects were allowed to take nutraceuticals other than n-3 PUFAs, the most common were vitamin, zinc or magnesium supplements. All subjects were instructed to take 4 capsules of the study product per day, added to their regular medication, during eight weeks. The daily dosage of the study product contained n-3 PUFAs equivalent to 2.2 g EPA, 400 mg DHA, 800 mg of other fatty acids and 10–24 mg tocopherol-rich extracts. The study product was provided by Bringwell AB that was acquired by Midsona AB during the course of the study. The content of the study product was analyzed, at three different time points during the production chain, for EPA/DHA content and lipid peroxidation, and the results of these analyses were found to be within reference limits.

All patients were treated with identical dose, open label n-3 PUFAs for 8 weeks. Symptoms were rated at baseline, week 4 and week 8 (end of study). We used a match-mismatch trial design which has been advocated by some as a means for advancing precision psychiatry (Miller et al., 2017; Miller and Pariante, 2020). In this trial design, patients are stratified by pre-treatment inflammation status, which is unknown to both the patient and the symptom rater during the course of the study. Our *a priori* hypothesis was that patients with high baseline inflammation would have a greater antidepressant response than those with low baseline inflammation. Compliance was evaluated using a diary in which patients noted all capsule intakes, and the remaining capsules were counted at the end of study.

2.3. CRP-assay

Plasma levels of hs-CRP was determined by an automated particle-based immunoassay (Aziz et al., 2003), adjusted to the international reference preparation ERM-DA-474/IFCC. The limit of quantification for the method was below 1 mg/L. During the course of the study, the hs-CRP assay at Skåne University Hospital was modified resulting in a lower detection limit for samples analyzed during the latter part of the study (0.3 mg/L instead of 0.6 mg/L). This did not affect our main analyses since the cut-off for defining the inflammation group was set to 1 mg/L or 3 mg/L. For correlation analyses involving CRP as a continuous variable, values of <0.6 mg/L were set to 0.3 mg/L and values <0.3 mg/L were set to 0.15 mg/L. In case we had several hs-CRP values taken at either baseline or week 8, a mean of these was calculated for correlation analyses.

2.4. Outcome measures

The primary outcome was baseline to week 8 change in HAMD-17 (Hamilton, 1960) scores. We also analyzed response and remission as binary variables. Response was defined as ≥50 % reduction in HAMD-17 score between baseline and week 8, and remission ≤ 7 on the HAMD-17 at week 8 (Frank et al., 1991; Bobo et al., 2016).

Secondary outcomes included change in: the self-rated version of Montgomery-Åsberg Depression Rating Scale (MADRS-S) (Svanborg and Asberg, 1994), a subscale of the Patient Health Questionnaire-9 (PHQ-9) (Kroenke et al., 2001) summarizing scores on symptom items previously associated with inflamed depression (Frank et al., 2021; Jokela et al., 2016; Fried et al., 2020; White et al., 2017) (items #3 sleep problems, #4 lack of energy, and #5 appetite-disturbance), the Generalized Anxiety Disorder 7-item scale (GAD-7) (Spitzer et al., 2006), the Snaith-

Hamilton Pleasure Scale (SHAPS) (Snaith et al., 1995), the Insomnia Severity Index (ISI) (Bastien et al., 2001), and the Fatigue Severity Scale (FSS) (Krupp et al., 1989).

2.5. Statistical analyses and power calculation

In the initial power calculation, the assumption was that we wanted to be able to demonstrate a between-group difference in change from baseline to week 8 in the primary outcome (HAMD-17 total score) of three points (Hengartner and Plöderl, 2022). To reject the null hypothesis that the population mean is equal between the groups (power = 0.80) we estimated that at least 45 patients with the outcome in each group are needed (two-tailed alpha = 0.05). The present match/mismatch study design did not include a placebo arm which may be considered a unique design with few previous comparable studies. Therefore, the sample size calculation was based on the best available information with regards to the variance of the primary outcome, derived from a similar patient group included in an RCT (Rapaport et al., 2016). After approximately 2/3 of the total sample had been included in the study, a statistician, not involved in any study procedures, analyzed the standard deviation of the primary outcome measure, and the number of subjects in each group. Based on this information, a new sample size calculation was performed, estimating that at least a total of 96 subjects were needed (power = 0.8, two-tailed alpha = 0.05).

Based on a previous study (Rapaport et al., 2016), the initial goal was to include at least 45 patients in the high inflammation group (initially defined as hs-CRP $\geq$ 3 mg/L) and at least 45 patients in the low inflammation group (initially defined as hs-CRP < 3 mg/L). There is, however, no consensus with regards to the most optimal hs-CRP cut-off to define "inflamed depression", and both $\geq$1 or $\geq$3 mg/L have been used in previous studies (Osimo et al., 2019). Hs-CRP values below 1 mg/L indicate a low risk for cardiovascular disease, 1–3 mg/L average risk, and values >3 mg/L high risk (Pearson et al., 2003; Ridker, 2003). After the interim check, it became clear that recruitment was lagging for the inflammation group and that the use of the pre-specified inflammation cut-off (hs-CRP $\geq$ 3 mg/L) would result in the inflammation group being substantially smaller than the non-inflammation group, and not reaching 45 individuals in each group which was pre-specified in the power calculation. After reviewing the literature and discussing with study collaborators we decided to analyze the data in two ways; with the inflammation group defined as $\geq$3 mg/L (*a priori* assumption) and as $\geq$1 mg/L (modified assumption after interim check). The 1 mg/L cut-off would result in more even groups consistent with the original power calculation.

In the initial recruitment phase, we obtained hs-CRP levels at two time-points, separated by at least one day and within approximately one week of starting the study to determine group status based on the initially proposed hs-CRP </$\geq$ 3 mg/L cut-off. In case of inconsistencies between the two values, a third hs-CRP was taken (in all cases except one, in whom the mean level of the two available hs-CRP values was used). Due to prolonged recruitment, the stratification process was modified during the course of the study to only require one hs-CRP value to determine group status. For stratification based on CRP </$\geq$1 mg/L, the first hs-CRP value obtained was used to implement a consistent methodology for the whole sample. Hereafter, groups are referred to as CRP3+ or CRP3− (CRP $\geq$ 3 mg/L or CRP < 3 mg/L according to our *a priori* assumption) or CRP1+ or CRP1− (CRP $\geq$ 1 mg/L or CRP < 1 mg/L according to the modified *post hoc* assumption).

Parametric tests were used since data were normally distributed, except analyses involving hs-CRP and BMI as continuous variables, then non-parametric tests were used. We used linear mixed model repeated measures to analyze change from baseline in outcome measures over the course of treatment. Inflammation group, treatment week (four and eight) and the interaction between inflammation group and treatment week were defined as fixed effects. Fisher's Exact Test was used for between-group comparisons of categorical variables. SAS Proc mixed was used and the unstructured covariance structure was chosen. The analyses were performed in IBM SPSS Statistics 28 Windows (IBM Corporation, Armonk, NY, USA) and SAS Enterprise Guide 8.3 (SAS Institute Inc., Cary, NC, USA). A significant p-value was defined at $\leq$0.05.

3. Results

3.1. Baseline characteristics and drop-outs

Of 536 patients screened, 126 were included in the study. Twenty-five of these were lost to follow-up between baseline and week 4, and seven patients were lost between week 4 and week 8, as presented in Fig. 1. 101 patients had at least one post-baseline follow-up visit and were hence included in the intention-to-treat analyses.

Demographic characteristics of the patients (divided by baseline hs-CRP-levels) are presented in Table 1 and Supplementary Table 1. Inflammation groups (CRP3+ and CRP1+) had significantly higher mean BMI compared to CRP3− and CRP1− respectively ($p < 0.001$). Baseline hs-CRP was significantly correlated with BMI (rho = 0.56, $p < 0.001$, n = 100). Selective serotonin reuptake inhibitors (SSRIs) were the most common class of antidepressants taken daily (64.4 % of the patients), as presented in Supplementary Table 2.

3.2. Change in outcome measures in patients with high and low hs-CRP

When the inflammation group cut-off of $\geq$3 mg/L on hs-CRP was used, there were no significant differences between groups in change in any of the outcome measures, either between baseline and week 4 or between baseline and week 8 (all $p > 0.13$), see Table 2 and Fig. 2a.

When the inflammation group cut-off of $\geq$1 mg/L on hs-CRP was used, there were several significant between-group differences in symptom change over the treatment course, i.e. the inflammation group improved more on the primary as well as several secondary outcome measures (Table 3).

Specifically, patients in CRP1+ showed a significantly greater decrease in total HAMD-17 score between baseline and week 8 compared to the CRP1− group ($p = 0.03$, Fig. 2B). The CRP1+ group also improved more on the HAMD-17 compared to the CRP1− group between baseline and week 4 ($p = 0.04$). Relative to the CRP1− group, the CRP + 1 group also showed a greater improvement on several secondary outcomes between baseline and week 8 including inflamed depression symptoms ($p = 0.05$), ISI ($p = 0.04$) and FSS ($p = 0.03$). The CRP1+ group improved more than the CRP1− group on SHAPS score between baseline and week 4 ($p = 0.008$), but this was not significant between baseline and week 8 ($p = 0.2$).

In exploratory analyses, change in outcome measures was compared between three groups; hs-CRP < 1 mg/L (n = 45), hs-CRP $\geq$ 1 and <3 mg/L (n = 27) and CRP $\geq$ 3 mg/L (n = 29). We found that the overall treatment effects were mainly driven by the hs-CRP 1–3 group that showed a significantly greater decrease in total HAMD-17 score compared to both the CRP < 1 group ($p < 0.001$) and the CRP $\geq$ 3 group ($p = 0.02$) with medium to large effect sizes. Similar results were seen for many of the secondary outcomes as shown in Supplementary Table 3.

3.3. Response and remission according to the HAMD-17

In total, 25 patients were responders and 18 of these remitters. In CRP3+ group, 34.6 % were responders and 19.2 % remitters compared to CRP3− group with 23.5 % responders and 19.1 % remitters ($p = 0.30$ and $p = 1.00$ respectively). In CRP1+ group, 38.5 % were responders and 28.8 % remitters compared to CRP1− group with 11.9 % responders and 7.1 % remitters ($p = 0.005$ and $p = 0.009$ respectively). Responders had signficantly higher baseline hs-CRP levels ($p = 0.017$) and BMI ($p = 0.018$) compared to non-responders. Remitters had signficantly higher baseline hs-CRP levels ($p = 0.037$) but not BMI ($p = 0.28$) compared to non-remitters. In correlation analyses, absolute change in HAMD-17

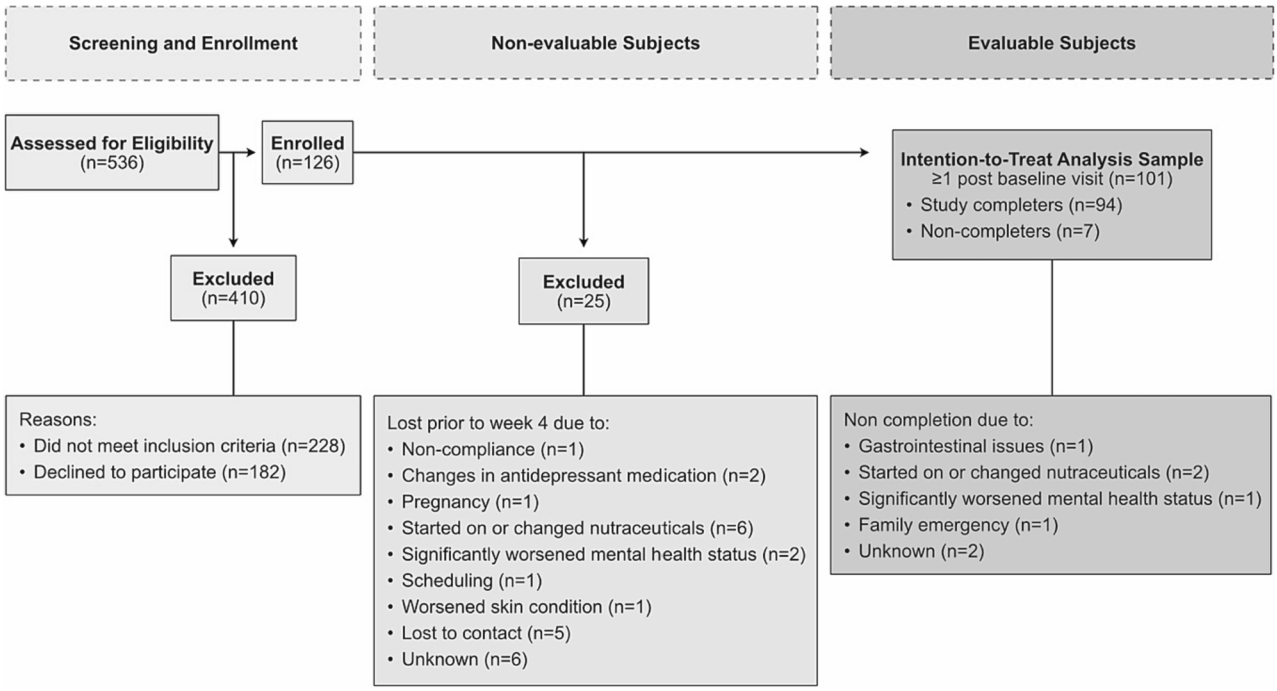


Fig. 1. Flow-chart of inclusion.

Table 1
Baseline demographic and clinical characteristics. Evaluable subjects (n = 101) with CRP < 3 mg/L (CRP3−) compared to CRP ≥ 3 mg/L (CRP3+) and patients with CRP < 1 mg/L (CRP1−) compared to CRP ≥ 1 mg/L (CRP1+).

	CRP3−	CRP3+	p	CRP1−	CRP1+	p
Evaluable subjects, *N*	72	29		45	56	
Age, *mean (s.d.) [range]*	39.4 (14.5) [18–72]	44.5 (13.6) [19–71]	0.11[a]	38.1 (14.9) [18–72]	43.0 (13.7) [19–71]	0.09[a]
Gender, *female, N (%)*	60 (83.3)	27 (93.1)	0.34[b]	39 (86.7)	48 (85.7)	1.00[b]
Smoking, *N (%)*	16 (22.9)	8 (27.6)	0.62[b]	10 (22.7)	14 (25.5)	0.82[b]
Snus use, *N (%)*	10 (14.7)	3 (10.3)	0.75[b]	5 (11.6)	8 (14.8)	0.77[b]
BMI[a], *Mean (s.d.) [range]*	24.2 (3.9) [18.4–36.8]	31.3 (6.8) [19.6–47.3]	<0.001*[a]	23.5 (3.9) [18.4–35.9]	28.5 (6.3) [19.5–47.3]	<0.001*[a]
Previous dep. *Mean (s.d.) [range]*	3.6 (3.9) [0–25]	4.6 (4.6) [0–20]	0.30[a]	3.7 (4.4) [0–25]	4.1 (3.9) [0–20]	0.66[a]
Previous ECT, *N (%)*	2 (3.0)	2 (6.9)	0.58[b]	2 (4.8)	2 (3.8)	1.00[b]
Previous suicide attempt, *N (%)*	13 (19.7)	4 (13.8)	0.57[b]	8 (19.0)	9 (17.0)	0.80[b]
HAMD-17 baseline, mean (s.d)	19.0 (3.3)	19.0 (3.7)	0.99[a]	19.5 (3.4)	18.6 (3.4)	0.18[a]
Non-n3 PUFA nutraceutical intake, N (%)	31 (43.1)	10 (34.5)	0.51[b]	22 (48.9)	19 (33.9)	0.16[b]

Abbreviations: CRP, C-reactive protein; BMI, body mass index; Previous dep; Previous depressive episodes (self-reported), ECT, electroconvulsive therapy. [a]BMI was calculated as: weight in kilograms divided by height in meters squared. Missing data; smoking (n = 2), snus use (n = 4), BMI (n = 1), Nr of depressive episodes (n = 2), previous ECT (n = 6) and previous suicide attempts (n = 6). * Significant at the p < 0.05 level. a) *t*-test, b) Fisher's Exact Test.

score between baseline and week 8 was not signficantly associated with baseline hs-CRP (rho = -0.08, p = 0.43) or BMI (rho = -0.16, p = 0.13).

3.4. Sensitivity analyses

In a sensitivity analysis, nine patients were excluded due to the following events during the study course: two patients suffered from mild COVID infection (n = 2); one patient underwent surgery; one patient had unsatisfactory compliance <85 % between baseline and week 4; two patients became aware of their baseline CRP level which was revealed at the week 8 visit, but the clinical assessor was still unaware of hs-CRP status at the time of evaluation; one patient was unable to come in for a physical visit at week 8 so this was done by phone; two patients started taking medications that may interfere with study results (methylphenidate and corticosteroids respectively). In the sensitivity analyses, there were still no significant group differences between the CRP3+ and CRP3− groups. The difference between CRP1+ and CRP1− groups in HAMD-17 was still significant between baseline and week 8 (p < 0.05, Hedge's g = 0.41), but not from BL to week 4 (p = 0.21). The CRP1+ group still had a superior response compared to the CRP1− group on secondary outcomes including inflamed depressive symptoms (p = 0.08, Hedge's g = 0.37), ISI (p = 0.10, Hedge's g = 0.34), and FSS (p = 0.13, Hedge's g = 0.33). These between-group differences in the secondary outcome measures were not statistically significant in the sensitivity analyses but effect sizes were similar compared to the intention-to-treat analyses (see Table 3).

3.5. Treatment-associated change in hs-CRP

Hs-CRP at week 8 was available for 92 subjects. In all subjects, there was no significant change in hs-CRP over the course of treatment (median, IQR at baseline: 1.15, 0.30–3.38 mg/L; median, IQR at week 8: 1.05, 0.32–4.03 mg/L, p = 0.98, Wilcoxon Signed Ranks Test). The CRP 1–3 mg/L group had a numerically greater decrease in hs-CRP than the CRP < 1 mg/L group but this did not reach statistical significance (p = 0.10, Mann-Whitney *U* test). Responders had a numerically greater

Table 2
Clinical outcomes in patients with CRP < 3 mg/L (CRP3−, n = 72) compared to CRP ≥ 3 mg/L (CRP3+, n = 29). Differences in symptom severity between groups from baseline (BL) to week 4 (W4) and week 8 (W8).

	Time	Group	Change with treatment		Group differences			Treatment effect
			LS-mean (SE)	95 % CI	Est (SE)	95 % CI	p	ES[a]
Primary outcome measure								
HAMD-17	W4-BL	CRP3−	−4.32 (0.45)	−5.22 to −3.42				
		CRP3+	−4.03 (0.71)	−5.45 to −2.62	−0.28 (0.84)	−1.96 to 1.39	0.74	−0.07
	W8-BL	CRP3−	−6.22 (0.52)	−7.25 to −5.19				
		CRP3+	−5.54 (0.93)	−7.40 to −3.69	−0.68 (1.07)	−2.80 to 1.45	0.53	−0.15
Secondary outcomes measures								
MADRS	W4-BL	CRP3−	−4.53 (0.73)	−5.97 to −3.08				
		CRP3+	−2.22 (1.33)	−4.86 to 0.43	−2.31 (1.52)	−5.33 to 0.71	0.13	−0.36
	W8-BL	CRP3−	−6.94 (0.86)	−8.66 to −5.22				
		CRP3+	−5.20 (1.31)	−7.81 to −2.60	−1.73 (1.57)	−4.86 to 1.39	0.27	−0.25
Inflamed Depression Symptoms	W4-BL	CRP3−	−1.20 (0.21)	−1.62 to −0.78				
		CRP3+	−1.27 (0.36)	−1.99 to −0.56	0.07 (0.42)	−0.76 to 0.90	0.87	0.04
	W8-BL	CRP3−	−1.92 (0.23)	−2.38 to −1.46				
		CRP3+	−1.68 (0.38)	−2.43 to −0.93	−0.24 (0.44)	−1.12 to 0.64	0.59	−0.13
GAD-7	W4-BL	CRP3−	−1.15 (0.49)	−2.13 to −0.17				
		CRP3+	−0.90 (0.80)	−2.49 to 0.69	−0.22 (0.94)	−2.12 to 1.62	0.79	−0.06
	W8-BL	CRP3−	−2.59 (0.53)	−3.64 to 1.53				
		CRP3+	−2.89 (0.69)	−4.27 to −1.52	0.31 (0.88)	−1.43 to 2.05	0.73	0.07
SHAPS	W4-BL	CRP3−	−3.62 (0.67)	−4.94 to −2.30				
		CRP3+	−3.07 (1.16)	−5.37 to −0.77	−0.55 (1.33)	−3.20 to 2.10	0.68	−0.09
	W8-BL	CRP3−	−5.59 (0.78)	−7.13 to −4.04				
		CRP3+	−4.77 (1.23)	−7.21 to −2.32	−0.82 (1.46)	−3.71 to 2.07	0.58	−0.13
ISI	W4-BL	CRP3−	−1.07 (0.48)	−2.01 to −0.12				
		CRP3+	−2.13 (0.77)	−3.66 to −0.61	1.07 (0.90)	−0.73 to 2.86	0.24	0.26
	W8-BL	CRP3−	−2.51 (0.52)	−3.54 to −1.48				
		CRP3+	−3.50 (0.93)	−5.36 to −1.65	0.99 (1.07)	−1.13 to 3.12	0.36	0.23
FSS	W4-BL	CRP3−	−1.00 (1.03)	−3.05 to 1.04				
		CRP3+	−1.42 (1.62)	−4.63 to 1.80	0.41 (1.92)	−3.41 to 4.23	0.83	0.05
	W8-BL	CRP3−	−3.02 (1.25)	−5.49 to −0.54				
		CRP3+	−4.09 (2.08)	−8.22 to 0.042	1.07 (2.43)	−3.75 to 5.90	0.66	0.1

Missing data: 7 subjects had missing data for all included rating scales at week 8. Abbreviations: Est; Estimation, CI; Confidence Interval, SE; Standard Error, ES; Effect Size, CRP; C-reactive protein, HAMD-17; Hamilton Depression Rating Scale 17-items, MADRS-S; self-rated version of Montgomery-Åsberg Depression Rating Scale, GAD-7; Generalized Anxiety Disorder 7-item scale, Inflamed Depression Symptoms; subscore of the Patient Health Questionnaire-9 including items #3 sleep problems, #4 lack of energy, and #5 appetite-disturbance, SHAPS; Snaith-Hamilton Pleasure Scale, ISI; Insomnia Severity Index, FSS; Fatigue Severity Scale. a) Hedge's g.

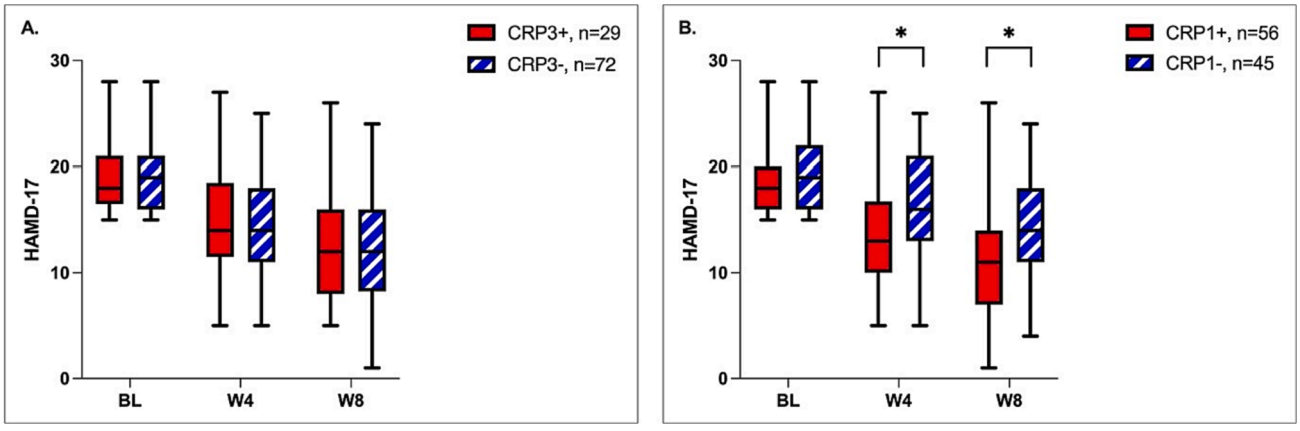


Fig. 2. Change in HAMD-17, comparisons between inflammation and non-inflammation groups A; subjects divided by hs-CRP</≥3 mg/L. B; subjects divided by hs-CRP</≥ 1 mg/L. Box and whiskers graph; boxes include second and third quartiles separated by median (solid line). Whiskers representing minimum to maximum. Missing data: A; w8 CRP3+ n = 3, CRP3− n = 4. B; w8 CRP1+ n = 4, CRP1 − n = 3.

decrease in hs-CRP than non-responders but this difference was not significant (p = 0.53, Mann-Whitney *U* test).

Absolute change in HAMD-17 did not correlate significantly with absolute change in hs-CRP over the course of treatment (rho = 0.08, p = 0.45, n = 91).

3.6. Adverse events

Omega-3 was overall well-tolerated by the study participants with few dropouts due to side effects (see Fig. 1) and no serious adverse events. Rates of reported adverse events are summarized in Supplementary Table 4.

Table 3
Clinical outcomes in patients with CRP < 1 mg/L (CRP1−, n = 45) compared to CRP ≥ 1 mg/L (CRP1+, n = 56). Differences in symptom severity between groups from baseline (BL) to week 4 (W4) and week 8 (W8).

	Time	Group	Change with treatment		Group differences			Treatment effect
			LS-mean (SE)	95 % CI	Est (SE)	95 % CI	p	ES[a]
Primary outcome measure								
HAMD-17	W4-BL	CRP1−	−3.36 (0.55)	−4.46 to −2.27				
		CRP1+	−4.94 (0.51)	−5.96 to −3.92	1.57 (0.76)	0.07–3.08	**0.04***	0.42
	W8-BL	CRP1−	−4.94 (0.64)	−6.22 to −3.67				
		CRP1+	−6.92 (0.63)	−8.16 to −5.68	1.98 (0.90)	0.20–3.77	**0.03***	0.45
Secondary outcome measures								
MADRS	W4-BL	CRP1−	−3.24 (0.89)	−5.00 to −1.47				
		CRP1+	−4.37 (0.94)	−6.23 to −2.51	1.13 (1.30)	−1.45 to 3.71	0.39	0.17
	W8-BL	CRP1−	−5.07 (1.02)	−7.10 to −3.04				
		CRP1+	−7.55 (1.01)	−9.55 to −5.54	2.48 (1.44)	−0.39 to 5.35	0.09	0.35
Inflamed Depression Symptoms	W4-BL	CRP1−	−0.87 (0.24)	−1.36 to −0.39				
		CRP1+	−1.50 (0.25)	−2.01 to −1.00	0.63 (0.35)	−0.07 to 1.33	0.08	0.35
	W8-BL	CRP1−	−1.43 (0.27)	−1.97 to −0.89				
		CRP1+	−2.20 (0.28)	−2.75 to −1.65	0.77 (0.39)	0.001–1.54	0.05	0.41
GAD-7	W4-BL	CRP1−	−0.64 (0.66)	−1.96 to 0.68				
		CRP1+	−1.43 (0.54)	−2.49 to −0.36	0.79 (0.86)	−0.91 to 2.49	0.36	0.19
	W8-BL	CRP1−	−2.41 (0.72)	−3.83 to −0.98				
		CRP1+	−2.89 (0.52)	−3.92 to −1.86	0.49 (0.89)	−1.28 to 2.25	0.59	0.12
SHAPS	W4-BL	CRP1−	−1.83 (0.65)	−3.12 to −0.55				
		CRP1+	−4.77 (0.87)	−6.49 to −3.05	2.94 (1.08)	0.78–5.09	**0.01***	0.52
	W8-BL	CRP1−	−4.41 (0.95)	−6.29 to −2.53				
		CRP1+	−6.10 (0.90)	−7.88 to −4.32	1.69 (1.31)	−0.91 to 4.29	0.20	0.27
ISI	W4-BL	CRP1−	−0.56 (0.60)	−1.75 to −0.62				
		CRP1+	−2.03 (0.54)	−3.10 to −0.96	1.46 (0.80)	−0.13 to 3.06	0.07	0.36
	W8-BL	CRP1−	−1.79 (0.59)	−2.96 to −0.63				
		CRP1+	−3.63 (0.67)	−4.95 to −2.30	1.83 (0.89)	0.07–3.60	**0.04***	0.42
FSS	W4-BL	CRP1−	0.29 (1.13)	−1.95 to 2.53				
		CRP1+	−2.26 (1.25)	−4.74 to 0.23	2.55 (1.68)	−0.80 to 5.89	0.13	0.30
	W8-BL	CRP1−	−0.80 (1.42)	−3.63 to 2.02				
		CRP1+	−5.27 (1.48)	−8.22 to −2.32	4.47 (2.06)	0.38–8.55	**0.03***	0.44

Missing data: 7 subjects had missing data for all included rating scales at week 8. Abbreviations: Est; Estimation, CI; Confidence Interval, SE; Standard Error, ES; Effect Size, CRP; C-reactive protein, HAMD-17; Hamilton Depression Rating Scale 17-items, MADRS-S; self-rated version of Montgomery-Åsberg Depression Rating Scale, GAD-7; Generalized Anxiety Disorder 7-item scale, Inflamed Depression Symptoms; subscore of the Patient Health Questionnaire-9 including items #3 sleep problems, #4 lack of energy, and #5 appetite-disturbance, SHAPS; Snaith-Hamilton Pleasure Scale, ISI; Insomnia Severity Index, FSS; Fatigue Severity Scale. a) Hedge's g
*Significant at the p < 0.05 level.

4. Discussion

In this study we found that patients with baseline hs-CRP ≥ 1 mg/L had a superior antidepressant response to n-3 PUFA EPA added to stable antidepressant treatment compared to those with hs-CRP < 1 mg/L. In addition to improvement in general depressive symptoms, add-on EPA had an effect on symptoms previously associated with inflamed depression such as fatigue and sleeping difficulties. When using a higher threshold for defining inflamed depression (hs-CRP ≥ 3 mg/L), however, significant treatment effects were not detected. Exploratory analyses showed that the treatment effect was mainly driven by the patients with hs-CRP between 1 mg/l and 3 mg/L.

4.1. Are n-3 PUFAs more effective in inflamed depression?

Many studies have investigated the efficacy of n-3 PUFA supplementation in depression in general, but results have been inconsistent and between-study heterogeneity relatively high (Appleton et al., 2021; Goh et al., 2021). Given the anti-inflammatory properties of n-3 PUFAs (Calder, 2006), their therapeutic potential has also been tested in several somatic conditions with inflammation as a central mechanism of disease, such as rheumatic diseases (Sigaux et al., 2022) and multiple sclerosis (AlAmmar et al., 2021). In the most recent Cochrane review and *meta*-analysis on the antidepressant efficacy of n-3 PUFAs, 28 RCTs (n = 1944) were included (Appleton et al., 2021). Effect sizes were small-to-modest with low quality of evidence. This *meta*-analysis did, however, not address the question if n-3 PUFAs are more efficacious in a subtype of depression with low-grade inflammation and associated metabolic alterations such as obesity, as has been suggested by several investigators (Rapaport et al., 2016; Su, 2015; Guu et al., 2019; Wu et al., 2023). In the present study, patients with elevated hs-CRP and better treatment response also had significantly elevated BMI and a trend toward higher age. This is consistent with a mechanistic link between obesity, inflammation and depression (Capuron et al., 2017), and inflammation being an important factor in the pathophysiology of old-age depression (Forbes et al., 2021). In a recent review, Wijaya et al. outlined the need for a multidimensional approach to classify inflamed depression (Wijaya et al., 2022). Consistent with this view, not only elevated inflammatory biomarkers, but also other clinical characteristics such as older age and higher BMI could be considered integral parts of the inflamed depression subtype. Another recent large-scale study reported a significant association between MDD polygenic risk score and hs-CRP, which became non-significant after adjusting for BMI and smoking (Pitharouli et al., 2021). These findings suggest that lifestyle and metabolic factors are key parts of the inflamed depression phenotype, and therefore we decided not to factor out BMI or age in our analyses of treatment effect.

Both clinical and experimental studies have shown that n-3 PUFAs have anti-inflammatory and neuroprotective properties that could underlie an antidepressant effect (Borsini et al., 2021). For instance, n-3 PUFAs can affect white blood cell chemotaxis, cell adhesion molecule expression, and immune response of microglia (Calder, 2013; Inoue et al., 2017). In addition to these well-established anti-inflammatory effects, n-3 PUFAs may also have beneficial effects on the hypothalamic–pituitary–adrenal axis and oxidative stress (Zhou et al., 2022) which could also mediate an antidepressant effect.

Consistent with our findings, Rapaport et al. showed that, while there was no robust antidepressant effect of n-3 PUFAs in MDD overall, there was however a large effect of EPA in patients with low-grade inflammation (Rapaport et al., 2016). Specifically, the reduction of depressive symptoms in the EPA-group was correlated with the number of baseline inflammation markers indicating 'high' inflammation (elevated hs-CRP, IL-6, IL-1ra, leptin or low adiponectin) (Rapaport et al., 2016). Patients with 4 or 5 elevated inflammatory markers at baseline demonstrated a treatment-associated decrease in depressive symptoms, relative to placebo, with large effect sizes. In a subsequent study from the same group, monotherapy EPA in different doses was compared to placebo in MDD patients, all with elevated hs-CRP and BMI (Mischoulon et al., 2022). Patients receiving the highest tested dose of EPA, 4 g/d, during the 12-week follow-up showed the greatest improvement. Also, improvement in depressive symptoms was associated with a decrease in CRP from baseline to week 12, in the 4 g/d group. Considering these results, optimal EPA dose for depression treatment might be higher than 2 g/d, which was the dose used in the current study. This may be especially true for patients with high inflammation (Mischoulon et al., 2022), and could help explain why we did not see a robust antidepressant effect in patients with ≥3 mg/L However, when the current study was planned, most n-3 PUFA studies in depression had used EPA supplementation doses ~1 g/d or less (Mocking et al., 2016; Bloch and Hannestad, 2012). Since a relatively higher dose of EPA was also predictive of a better response (Mocking et al., 2016), we decided to use 2 g EPA per day added to a stable antidepressant medication in our study. Yet, future studies need to investigate which EPA dose is optimal in inflamed depression and if this differs between monotherapy vs. add-on studies.

4.2. Baseline hs-CRP as a biomarker for treatment response

Mounting evidence suggest that a subset of depressed patients shows signs of concurrent low-grade inflammation (Osimo et al., 2019), and these patients are more likely to be obese, have somatic co-morbidity and often show poorer response to regular antidepressants (Miller et al., 2017; Strawbridge et al., 2015). There is currently no consensus with regards to which biomarkers and cut-offs that should be used to best define inflamed depression (Young et al., 2016). Peripheral hs-CRP has been proposed as a candidate biomarker for this purpose, but most previous studies have relied on cut-offs established for risk stratification in cardiovascular medicine (Pearson et al., 2003). The translation of these cut-offs into depression research should therefore be considered arbitrary to some degree, and more studies are needed to determine most optimal cut-off for predicting relevant clinical outcomes. In a *meta*-analysis including >13 000 depressed patients, 58 % had CRP > 1 mg/L and 27 % had CRP > 3 mg/L (Osimo et al., 2019), which is similar to how CRP-levels were distributed in the present study. Hs-CRP cut-offs of >1 mg/L (Nettis et al., 2021; Pan et al., 2022; Jha et al., 2017), >3 mg/L (Suneson et al., 2023; Rapaport et al., 2016; Felger et al., 2020), and >5 mg/L (McIntyre et al., 2019; Raison et al., 2013), have all been used to identify depressed subjects with low-grade inflammation. In the present study we chose to stratify on moderate (≥1 mg/L) and high (≥3 mg/L) elevations of hs-CRP. This is consistent with hs-CRP cut-offs suggested by the Centers for Disease Control and Prevention (CDC) indicating 'low' (<1 mg/L) 'average' (1–3 mg/L) and 'high' (>3 mg/L) risk of cardiovascular disease. Furthermore, these cut-offs are also consistent with approximations of hs-CRP tertiles in the general adult population (Pearson et al., 2003). As hs-CRP above 3 mg/L was associated with a robust antidepressant effect of EPA in the study by Rapaport et al. (Rapaport et al., 2016), we chose this as our primary cut-off, and a lower cut-off of 1 mg/L was added in secondary analyses. This is consistent with several other studies also using the 1 mg/L cut-off to investigate low-grade inflammation in depression (Nettis et al., 2021; Pan et al., 2022; Jha et al., 2017; Sforzini et al., 2023), and for sample enrichment when testing the antidepressant efficacy of anti-inflammatory drugs such as minocycline (Nettis et al., 2021). In general population samples, CRP levels >1 mg/L have been associated with increased odds for antidepressant use and prescription (Wium-Andersen et al., 2013), as well as ongoing and recurrent depression in male subjects (Liukkonen et al., 2006). Moreover, elevations of hs-CRP > 1 mg/L have been shown to predict poorer treatment response to SSRIs compared to other classes of antidepressants affecting other monoaminergic systems (Pan et al., 2022; Jha et al., 2017; Uher et al., 2014). Hence, even moderate pre-treatment elevations of hs-CRP may be relevant to guide and predict treatment response in MDD.

While our *a priori* chosen cut-off for hs-CRP (3 mg/L) did not predict a superior antidepressant effect to EPA, additional analyses using 1 mg/L as hs-CRP cut-off did. Specifically, we found a significant decrease of HAMD-17 total score, as well as significantly more responders and remitters, in patients with baseline hs-CRP levels above 1 mg/L compared to below 1 mg/L. Moreover, exploratory analyses revealed that the antidepressant affect was strongest in those patients with hs-CRP between 1 mg/L and 3 mg/L. Our findings that EPA showed an antidepressant effect even at moderately elevated pre-treatment hs-CRP levels suggest that this biomarker may be even more sensitive to predicting treatment response than originally expected. These findings raise the possibility that 1 mg/L may be a better cut-off than 3 mg/L to predict n-3 PUFA antidepressant treatment response, although larger scale studies with diverse populations are needed to confirm this.

In a subset of the patients, we did not observe a significant change in hs-CRP over the course of treatment, nor did change in hs-CRP correlate significantly with treatment response. Previous studies investigating if n-3 PUFAs reduce CRP have shown conflicting results (Mischoulon et al., 2022; Dezfouli et al., 2020; Elisia et al., 2022; Mori et al., 2003; Mortazavi et al., 2018; Natto et al., 2019; Tayyebi-Khosroshahi et al., 2012), and this effect may depend on a number of factors such as dose, duration, baseline inflammation status and type of illness to name a few. Hs-CRP is a non-specific inflammatory marker that is not able to account for all the biological complexities thought to underlie inflamed depression. For instance, Sforzini et al. recently showed that several immune-related genes were upregulated in depression compared to controls, regardless of high or low hs-CRP-levels (Sforzini et al., 2023). Nevertheless, hs-CRP has several advantages as a potential clinical biomarker to aid selection of antidepressant treatment. It is an accessible biomarker that can be assayed in a standardized way and with quick and relatively robust results across labs. Moreover, hs-CRP levels are less reactive to unspecific acute stress (Marsland et al., 2017) than some other inflammatory markers. Furthermore, some studies have shown that peripheral hs-CRP correlates with several meaningful central indices of inflamed depression such as lower corticostriatal connectivity (Felger et al., 2016), higher basal ganglia glutamate (Haroon et al., 2016), and hs-CRP in cerebrospinal fluid (Felger et al., 2020; Lindqvist et al., 2013). Accordingly, we believe that hs-CRP is a useful marker to stratify patients based on inflammation status as was used in our study design, although we acknowledge that it does not capture all the complexities thought to be involved in inflamed depression. Future personalized treatment studies should also consider complementary strategies including more multidimensional and possibly longitudinal approaches than only hs-CRP stratification. For example, several previous studies have used enrichment designs (Mischoulon et al., 2022; Nettis et al., 2021; McIntyre et al., 2019) considering combinations of biomarkers and/or clinical characteristics to define inflamed and/or immunometabolic depression to test anti-inflammatory compounds in such subsets of patients. Future studies enriched for inflamed depression and trials with designs similar to ours may jointly be of importance to advance precision psychiatry.

4.3. Do n-3 PUFAs have a specific effect on inflammatory depressive symptoms?

Accumulating evidence points toward specific depressive symptom dimensions being more strongly associated with elevated peripheral

inflammatory markers (Frank et al., 2021; Suneson et al., 2023; Jokela et al., 2016; Fried et al., 2020; White et al., 2017). In addition to an overall antidepressant effect, we found that moderately elevated hs-CRP (≥1 mg/L) predicted improvement in symptom dimensions previously related to inflammation, including sleep problems, fatigue, lack of energy, anhedonia and appetite disturbances (Frank et al., 2021; Suneson et al., 2023). Interestingly, these symptom dimensions overlap with key components of sickness-behavior, a term used to describe alterations of mood and behavior following immune-challenges such as infections (Dantzer et al., 2008; Turkheimer et al., 2023). Our findings are in line with results from several large-scale general population studies in which both fatigue, lack of energy and sleep problems have been positively correlated to high CRP-levels (Frank et al., 2021; Jokela et al., 2016; White et al., 2017). In accordance, a recent study on young adults with MDD found elevated CRP-levels among patients with more pronounced neurovegetative symptomatology (including tiredness, increased appetite and weight gain) (Franklyn et al., 2022). In a cross-sectional study of difficult-to-treat depression, we have previously shown that patients with higher CRP-levels had more severe symptoms of fatigue, changes in sleep and appetite, inability to feel and lassitude, but not overall depression severity (Suneson et al., 2023). While the current study found significant treatment effects of EPA on putatively "inflamed depressive symptoms", some of these findings missed statistical significance in sensitivity analyses, excluding patients with various protocol deviations. However, as effect sizes were comparable, we hypothesize that this might have been a power issue. Overall, we claim that these findings should encourage future studies on EPA supplementation and other targeted interventions for inflamed depression to include assessment of specific symptom dimensions of MDD, rather than to solely evaluate overall depression severity.

4.4. Strengths and limitations

To our knowledge, this is one of the first antidepressant trials to use a match-mismatch design without a placebo arm. This type of novel study design has been advocated as a means to study the specificity of any antidepressant effect of an anti-inflammatory compound in a subgroup of patients with inflammation (Miller et al., 2017). Ultimately, this study design could help answer the relevant clinical question if a biomarker, or any other measurement, can be used *a priori* to predict the chance of treatment success in a given depressed individual, which ideally could pave the way for precision psychiatry. Our findings suggest that n-3 PUFA supplementation may be a useful treatment option for some patients. Patients with hs-CRP ≥ 1 had a mean change in HAMD-17 of approximately 7 points which is generally considered clinically meaningful (Rush et al., 2021) and the effect size relative to the CRP < 1 group was 0.45 which is considered a medium size effect.

Our study also comes with several notable limitations. For instance, we were not able to control for dietary intake or change in health behaviors (including physical activity) the weeks before or during the study. All subjects agreed, however, not to significantly modify their diet during the course of the study. While hs-CRP holds many preferable properties for a biomarker, as discussed above, results from one single measurement (as used in later stages of this study) may not capture temporary fluctuations in hs-CRP-levels. Such temporary fluctuations also includes diurnal variation which may be of significance according to some (Bogaty et al., 2013; Koc et al., 2010), but not all (Meier-Ewert et al., 2001), previous studies. It is therefore a potential limitation that we did not sample blood for hs-CRP analyses at a consistent time of day. The study was originally powered to detect a between-group difference in change in the primary outcome measure (HAMD-17 total score) and was not based on the secondary outcomes. Moreover, we did not adjust the alpha level when evaluating the antidepressant effect of n-3 PUFAs on the secondary outcomes. Therefore, these secondary analyses should be considered preliminary and in need of replication. We conducted an interim check after approximately 2/3 of the patients had been included. Although reviewing data during the study may come with certain risks, none of the research physicians that would assess symptoms in study participants enrolled after the interim check were involved in these procedures, thus we do not believe that this influenced our results. Another potential limitation of our study was that we did not include a placebo arm, instead all patients were given identical treatment and raters and patients were blind for inflammation status. Future studies should consider including placebo arms in both inflammation and non-inflammation groups to further validate these findings. Also, since our study included patients with mild-to-moderate depression and excluded those with autoimmune disorders and other inflammatory disorders, our results may not be generalizable to depressed populations with more severe illness and somatic co-morbidities.

4.5. Conclusion

We here report that EPA supplementation has a greater antidepressant effect in patients with moderately elevated hs-CRP compared to patients with lower hs-CRP. These findings add to the mounting evidence that stratification of MDD subgroups based on inflammatory markers may be relevant to predict response to anti-inflammatory interventions. Although more studies are needed to confirm the most optimal cut-off to define such subgroups, this and other similar studies have the potential to facilitate the development of targeted interventions and thereby break ground for precision medicine in psychiatry (Drevets et al., 2022).

Funding

The study was funded by the Swedish Research Council (grant number 2020-01428), Swedish governmental funding of clinical research (ALF), the OM Persson Foundation, the Ellen and Henrik Sjöbring Foundation, the Söderström Königska Foundation, the Lions Research Foundation, the Swedish Society of Medicine, Skåne University Hospital Donations, the Per-Eric and Ulla Schybergs Foundation, and the Professor Bror Gadelius memorial fund.

CRediT authorship contribution statement

Klara Suneson: Conceptualization, Data curation, Formal analysis, Investigation, Visualization, Writing – original draft. **Gustav Söderberg Veibäck:** Data curation, Investigation, Writing – review & editing. **Jesper Lindahl:** Data curation, Investigation, Writing – review & editing. **Johanna Tjernberg:** Data curation, Investigation, Project administration, Writing – review & editing. **Darya Ståhl:** Data curation, Investigation, Project administration, Writing – review & editing. **Simon Ventorp:** Investigation, Project administration, Writing – review & editing. **Filip Ängeby:** Investigation, Writing – review & editing. **Karl Lundblad:** Investigation, Project administration, Writing – review & editing. **Owen M. Wolkowitz:** Funding acquisition, Supervision, Writing – review & editing. **Daniel Lindqvist:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Supervision, Writing – original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

In accordance with the Swedish Public Access to Information and Secrecy Act, data cannot be made freely available. Requests of data can be sent to registrator@lu.se.

Acknowledgements

Authors would like to extend our gratitude to all of those who have participated in and worked with the current study. Among our colleagues who have contributed are Sofie Manninen, Lars-Olof Angmo, Agnes Eriksson and Johan Olsson. For support in statistical analyses and preparation of this paper we thank Helene Jacobsson and Magnus Förnvik. We thank Bringwell AB and Midsona AB for donating the study intervention.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.bbi.2024.02.029.

References

AlAmmar, W.A., Albeesh, F.H., Ibrahim, L.M., Algindan, Y.Y., Yamani, L.Z., Khattab, R. Y., 2021. Effect of omega-3 fatty acids and fish oil supplementation on multiple sclerosis: a systematic review. Nutr. Neurosci. 24 (7), 569–579.

APA, 2013. Diagnostic and statistical manual of mental disorders (DSM-5®). American Psychiatric Publications.

Appleton, K.M., Voyias, P.D., Sallis, H.M., Dawson, S., Ness, A.R., Churchill, R., et al., 2021. Omega-3 fatty acids for depression in adults. Cochrane Database Syst. Rev. 11, CD004692.

Aziz, N., Fahey, J.L., Detels, R., Butch, A.W., 2003. Analytical performance of a highly sensitive C-reactive protein-based immunoassay and the effects of laboratory variables on levels of protein in blood. Clin. Diagn. Lab. Immunol. 10 (4), 652–657.

Bastien, C.H., Vallières, A., Morin, C.M., 2001. Validation of the insomnia severity index as an outcome measure for insomnia research. Sleep Med. 2 (4), 297–307.

Bloch, M.H., Hannestad, J., 2012. Omega-3 fatty acids for the treatment of depression: systematic review and meta-analysis. Mol. Psychiatry 17 (12), 1272–1282.

Bobo, W.V., Anglero, G.C., Jenkins, G., Hall-Flavin, D.K., Weinshilboum, R., Biernacka, J. M., 2016. Validation of the 17-item Hamilton depression rating scale definition of response for adults with major depressive disorder using equipercentile linking to clinical global impression scale ratings: analysis of pharmacogenomic research network antidepressant medication pharmacogenomic study (PGRN-AMPS) data. Hum. Psychopharmacol. 31 (3), 185–192.

Bogaty, P., Dagenais, G.R., Joseph, L., Boyer, L., Leblanc, A., Bélisle, P., et al., 2013. Time variability of C-reactive protein: implications for clinical risk stratification. PLoS One 8 (4), e60759.

Borsini, A., Nicolaou, A., Camacho-Muñoz, D., Kendall, A.C., Di Benedetto, M.G., Giacobbe, J., et al., 2021. Omega-3 polyunsaturated fatty acids protect against inflammation through production of LOX and CYP450 lipid mediators: relevance for major depression and for human hippocampal neurogenesis. Mol. Psychiatry 26 (11), 6773–6788.

Busner, J., Targum, S.D., 2007. The clinical global impressions scale: applying a research tool in clinical practice. Psychiatry (Edgmont) 4 (7), 28–37.

Calder, P.C., 2006. n-3 polyunsaturated fatty acids, inflammation, and inflammatory diseases. Am. J. Clin. Nutr. 83 (6 Suppl), 1505S–S1519.

Calder, P.C., 2013. Omega-3 polyunsaturated fatty acids and inflammatory processes: nutrition or pharmacology? Br. J. Clin. Pharmacol. 75 (3), 645–662.

Calder, P.C., 2017. Omega-3 fatty acids and inflammatory processes: from molecules to man. Biochem. Soc. Trans. 45 (5), 1105–1115.

Capuron, L., Lasselin, J., Castanon, N., 2017. Role of adiposity-driven inflammation in depressive morbidity. Neuropsychopharmacology 42 (1), 115–128.

Dantzer, R., O'Connor, J.C., Freund, G.G., Johnson, R.W., Kelley, K.W., 2008. From inflammation to sickness and depression: when the immune system subjugates the brain. Nat. Rev. Neurosci. 9 (1), 46–56.

Dezfouli, M., Moeinzadeh, F., Taheri, S., Feizi, A., 2020. The effect of omega-3 supplementation on serum levels of inflammatory biomarkers and albumin in hemodialysis patients: a systematic review and meta-analysis. J. Ren. Nutr. 30 (3), 182–188.

Drevets, W.C., Wittenberg, G.M., Bullmore, E.T., Manji, H.K., 2022. Immune targets for therapeutic development in depression: towards precision medicine. Nat. Rev. Drug Discov. 21 (3), 224–244.

Elisia, I., Yeung, M., Kowalski, S., Wong, J., Rafiei, H., Dyer, R.A., et al., 2022. Omega 3 supplementation reduces C-reactive protein, prostaglandin E(2) and the granulocyte/lymphocyte ratio in heavy smokers: an open-label randomized crossover trial. Front. Nutr. 9, 1051418.

Felger, J.C., Li, Z., Haroon, E., Woolwine, B.J., Jung, M.Y., Hu, X., et al., 2016. Inflammation is associated with decreased functional connectivity within corticostriatal reward circuitry in depression. Mol. Psychiatry 21 (10), 1358–1365.

Felger, J.C., Haroon, E., Patel, T.A., Goldsmith, D.R., Wommack, E.C., Woolwine, B.J., et al., 2020. What does plasma CRP tell us about peripheral and central inflammation in depression? Mol. Psychiatry 25 (6), 1301–1311.

Forbes, M.P., O'Neil, A., Lane, M., Agustini, B., Myles, N., Berk, M., 2021. Major depressive disorder in older patients as an inflammatory disorder: implications for the pharmacological management of geriatric depression. Drugs Aging 38 (6), 451–467.

Frank, P., Jokela, M., Batty, G.D., Cadar, D., Steptoe, A., Kivimäki, M., 2021. Association between systemic inflammation and individual symptoms of depression: a pooled analysis of 15 population-based cohort studies. Am. J. Psychiatry 178 (12), 1107–1118.

Frank, E., Prien, R.F., Jarrett, R.B., Keller, M.B., Kupfer, D.J., Lavori, P.W., et al., 1991. Conceptualization and rationale for consensus definitions of terms in major depressive disorder. Remission, recovery, relapse, and recurrence. Arch. Gen. Psychiatry 48 (9), 851–855.

Franklyn, S.I., Stewart, J., Beaurepaire, C., Thaw, E., McQuaid, R.J., 2022. Developing symptom clusters: linking inflammatory biomarkers to depressive symptom profiles. Transl. Psychiatry 12 (1), 133.

Fried, E.I., von Stockert, S., Haslbeck, J.M.B., Lamers, F., Schoevers, R.A., Penninx, B.W. J.H., 2020. Using network analysis to examine links between individual depressive symptoms, inflammatory markers, and covariates. Psychol. Med. 50 (16), 2682–2690.

Goh, K.K., Chen, C.Y., Chen, C.H., Lu, M.L., 2021. Effects of omega-3 polyunsaturated fatty acids supplements on psychopathology and metabolic parameters in schizophrenia: a meta-analysis of randomized controlled trials. J. Psychopharmacol. 35 (3), 221–235.

Guu, T.W., Mischoulon, D., Sarris, J., Hibbeln, J., McNamara, R.K., Hamazaki, K., et al., 2019. International society for nutritional psychiatry research practice guidelines for omega-3 fatty acids in the treatment of major depressive disorder. Psychother. Psychosom. 88 (5), 263–273.

Hamilton, M., 1960. A rating scale for depression. J. Neurol. Neurosurg. Psychiatry 23, 56–62.

Haroon, E., Fleischer, C.C., Felger, J.C., Chen, X., Woolwine, B.J., Patel, T., et al., 2016. Conceptual convergence: increased inflammation is associated with increased basal ganglia glutamate in patients with major depression. Mol. Psychiatry 21 (10), 1351–1357.

Haroon, E., Welle, J.R., Woolwine, B.J., Goldsmith, D.R., Baer, W., Patel, T., et al., 2020. Associations among peripheral and central kynurenine pathway metabolites and inflammation in depression. Neuropsychopharmacology 45 (6), 998–1007.

Hengartner, M.P., Plöderl, M., 2022. Estimates of the minimal important difference to evaluate the clinical significance of antidepressants in the acute treatment of moderate-to-severe depression. BMJ Evid Based Med. 27 (2), 69–73.

Inoue, T., Tanaka, M., Masuda, S., Ohue-Kitano, R., Yamakage, H., Muranaka, K., et al., 2017. Omega-3 polyunsaturated fatty acids suppress the inflammatory responses of lipopolysaccharide-stimulated mouse microglia by activating SIRT1 pathways. Biochim. Biophys. Acta Mol. Cell Biol. Lipids 1862 (5), 552–560.

Jha, M.K., Minhajuddin, A., Gadad, B.S., Greer, T., Grannemann, B., Soyombo, A., et al., 2017. Can C-reactive protein inform antidepressant medication selection in depressed outpatients? Findings from the CO-MED trial. Psychoneuroendocrinology 78, 105–113.

Jokela, M., Virtanen, M., Batty, G.D., Kivimaki, M., 2016. Inflammation and specific symptoms of depression. JAMA Psychiat. 73 (1), 87–88.

Koc, M., Karaarslan, O., Abali, G., Batur, M.K., 2010. Variation in high-sensitivity C-reactive protein levels over 24 hours in patients with stable coronary artery disease. Tex. Heart Inst. J. 37 (1), 42–48.

Kroenke, K., Spitzer, R.L., Williams, J.B., 2001. The PHQ-9: validity of a brief depression severity measure. J. Gen. Intern. Med. 16 (9), 606–613.

Krupp, L.B., LaRocca, N.G., Muir-Nash, J., Steinberg, A.D., 1989. The fatigue severity scale. application to patients with multiple sclerosis and systemic lupus erythematosus. Arch. Neurol. 46 (10), 1121–1123.

Liao, Y., Xie, B., Zhang, H., He, Q., Guo, L., Subramanieapillai, M., et al., 2019. Efficacy of omega-3 PUFAs in depression: a meta-analysis. Transl. Psychiatry 9 (1), 190.

Lindqvist, D., Hall, S., Surova, Y., Nielsen, H.M., Janelidze, S., Brundin, L., et al., 2013. Cerebrospinal fluid inflammatory markers in Parkinson's disease–associations with depression, fatigue, and cognitive impairment. Brain Behav. Immun. 33, 183–189.

Liukkonen, T., Silvennoinen-Kassinen, S., Jokelainen, J., Räsänen, P., Leinonen, M., Meyer-Rochow, V.B., et al., 2006. The association between C-reactive protein levels and depression: results from the northern Finland 1966 birth cohort study. Biol. Psychiatry 60 (8), 825–830.

Lynall, M.E., Turner, L., Bhatti, J., Cavanagh, J., de Boer, P., Mondelli, V., et al., 2020. Peripheral blood cell-stratified subgroups of inflamed depression. Biol. Psychiatry 88 (2), 185–196.

Marsland, A.L., Walsh, C., Lockwood, K., John-Henderson, N.A., 2017. The effects of acute psychological stress on circulating and stimulated inflammatory markers: a systematic review and meta-analysis. Brain Behav. Immun. 64, 208–219.

McIntyre, R.S., Subramaniapillai, M., Lee, Y., Pan, Z., Carmona, N.E., Shekotikhina, M., et al., 2019. Efficacy of adjunctive infliximab vs placebo in the treatment of adults with bipolar I/II depression: a randomized clinical trial. JAMA Psychiat. 76 (8), 783–790.

Meier-Ewert, H.K., Ridker, P.M., Rifai, N., Price, N., Dinges, D.F., Mullington, J.M., 2001. Absence of diurnal variation of C-reactive protein concentrations in healthy human subjects. Clin. Chem. 47 (3), 426–430.

Miller, A.H., Haroon, E., Felger, J.C., 2017. Therapeutic implications of brain-immune interactions: treatment in translation. Neuropsychopharmacology 42 (1), 334–359.

Miller, A.H., Pariante, C.M., 2020. Trial failures of anti-inflammatory drugs in depression. Lancet Psychiatry 7 (10), 837.

Mischoulon, D., Dunlop, B.W., Kinkead, B., Schettler, P.J., Lamon-Fava, S., Rakofsky, J. J., et al., 2022. Omega-3 fatty acids for major depressive disorder with high inflammation: a randomized dose-finding clinical trial. J. Clin. Psychiatry 83 (5).

Mocking, R.J., Harmsen, I., Assies, J., Koeter, M.W., Ruhe, H.G., Schene, A.H., 2016. Meta-analysis and meta-regression of omega-3 polyunsaturated fatty acid supplementation for major depressive disorder. Transl. Psychiatry 6 (3), e756.

Mori, T.A., Woodman, R.J., Burke, V., Puddey, I.B., Croft, K.D., Beilin, L.J., 2003. Effect of eicosapentaenoic acid and docosahexaenoic acid on oxidative stress and inflammatory markers in treated-hypertensive type 2 diabetic subjects. Free Radic. Biol. Med. 35 (7), 772–781.
Mortazavi, A., Nematipoor, E., Djalali, M., Keshavarz, S.A., Samavat, S., Zarei, M., et al., 2018. The effect of omega-3 fatty acids on serum apelin levels in cardiovascular disease: a randomized, double-blind, placebo-controlled trial. Rep. Biochem. Mol. Biol. 7 (1), 59–66.
Natto, Z.S., Yaghmoor, W., Alshaeri, H.K., Van Dyke, T.E., 2019. Omega-3 fatty acids effects on inflammatory biomarkers and lipid profiles among diabetic and cardiovascular disease patients: a systematic review and meta-analysis. Sci. Rep. 9 (1), 18867.
Nettis, M.A., Lombardo, G., Hastings, C., Zajkowska, Z., Mariani, N., Nikkheslat, N., et al., 2021. Augmentation therapy with minocycline in treatment-resistant depression patients with low-grade peripheral inflammation: results from a double-blind randomised clinical trial. Neuropsychopharmacology 46 (5), 939–948.
Osimo, E.F., Baxter, L.J., Lewis, G., Jones, P.B., Khandaker, G.M., 2019. Prevalence of low-grade inflammation in depression: a systematic review and meta-analysis of CRP levels. Psychol. Med. 49 (12), 1958–1970.
Pan, Y., Luo, R., Zhang, S., Liu, Y., Wang, Y., Feng, S., et al., 2022. C-reactive protein could predict the efficacy of SSRIs in clinical practice: a cohort study of large samples in the real world. J. Affect. Disord. 313, 251–259.
Pearson, T.A., Mensah, G.A., Alexander, R.W., Anderson, J.L., Cannon 3rd, R.O., Criqui, M., et al., 2003. Markers of inflammation and cardiovascular disease: application to clinical and public health practice: a statement for healthcare professionals from the Centers for Disease Control and Prevention and the American Heart Association. Circulation 107 (3), 499–511.
Pitharouli, M.C., Hagenaars, S.P., Glanville, K.P., Coleman, J.R.I., Hotopf, M., Lewis, C. M., et al., 2021. Elevated C-reactive protein in patients with depression, independent of genetic, health, and psychosocial factors: results from the UK biobank. Am. J. Psychiatry 178 (6), 522–529.
Raison, C.L., Miller, A.H., 2011. Is depression an inflammatory disorder? Curr. Psychiatry Rep. 13 (6), 467–475.
Raison, C.L., Rutherford, R.E., Woolwine, B.J., Shuo, C., Schettler, P., Drake, D.F., et al., 2013. A randomized controlled trial of the tumor necrosis factor antagonist infliximab for treatment-resistant depression: the role of baseline inflammatory biomarkers. JAMA Psychiat. 70 (1), 31–41.
Rapaport, M.H., Nierenberg, A.A., Schettler, P.J., Kinkead, B., Cardoos, A., Walker, R., et al., 2016. Inflammation as a predictive biomarker for response to omega-3 fatty acids in major depressive disorder: a proof-of-concept study. Mol. Psychiatry 21 (1), 71–79.
Ridker, P.M., 2003. Clinical application of C-reactive protein for cardiovascular disease detection and prevention. Circulation 107 (3), 363–369.
Rush, A.J., South, C., Jain, S., Agha, R., Zhang, M., Shrestha, S., et al., 2021. Clinically significant changes in the 17- and 6-item Hamilton rating scales for depression: a STAR*D report. Neuropsychiatr. Dis. Treat. 17, 2333–2345.
Sforzini, L., Cattaneo, A., Ferrari, C., Turner, L., Mariani, N., Enache, D., et al., 2023. Higher immune-related gene expression in major depression is independent of CRP levels: results from the BIODEP study. Transl. Psychiatry 13 (1), 185.
Sigaux, J., Mathieu, S., Nguyen, Y., Sanchez, P., Letarouilly, J.G., Soubrier, M., et al., 2022. Impact of type and dose of oral polyunsaturated fatty acid supplementation on disease activity in inflammatory rheumatic diseases: a systematic literature review and meta-analysis. Arthritis Res. Ther. 24 (1), 100.
Snaith, R.P., Hamilton, M., Morley, S., Humayan, A., Hargreaves, D., Trigwell, P., 1995. A scale for the assessment of hedonic tone the snaith-Hamilton pleasure scale. Br. J. Psychiatry 167 (1), 99–103.
Spitzer, R.L., Kroenke, K., Williams, J.B., Löwe, B., 2006. A brief measure for assessing generalized anxiety disorder: the GAD-7. Arch. Intern. Med. 166 (10), 1092–1097.
Strawbridge, R., Arnone, D., Danese, A., Papadopoulos, A., Herane Vives, A., Cleare, A.J., 2015. Inflammation and clinical response to treatment in depression: a meta-analysis. Eur. Neuropsychopharmacol. 25 (10), 1532–1543.
Su, K.P., 2015. Personalized medicine with Omega-3 fatty acids for depression in children and pregnant women and depression associated with inflammation. J. Clin. Psychiatry 76 (11), e1476–e1477.
Su, K.P., Lai, H.C., Yang, H.T., Su, W.P., Peng, C.Y., Chang, J.P., et al., 2014. Omega-3 fatty acids in the prevention of interferon-alpha-induced depression: results from a randomized, controlled trial. Biol. Psychiatry 76 (7), 559–566.
Suneson, K., Lindahl, J., Chamli Harsmar, S., Soderberg, G., Lindqvist, D., 2021. Inflammatory depression-mechanisms and non-pharmacological interventions. Int. J. Mol. Sci. 22 (4).
Suneson, K., Angeby, F., Lindahl, J., Soderberg, G., Tjernberg, J., Lindqvist, D., 2022. Efficacy of eicosapentaenoic acid in inflammatory depression: study protocol for a match-mismatch trial. BMC Psychiatry 22 (1), 801.
Suneson, K., Grudet, C., Ventorp, F., Malm, J., Asp, M., Westrin, Å., et al., 2023. An inflamed subtype of difficult-to-treat depression. Prog. Neuropsychopharmacol. Biol. Psychiatry 125, 110763.
Svanborg, P., Asberg, M., 1994. A new self-rating scale for depression and anxiety states based on the comprehensive psychopathological rating scale. Acta Psychiatr. Scand. 89 (1), 21–28.
Tayyebi-Khosroshahi, H., Houshyar, J., Dehgan-Hesari, R., Alikhah, H., Vatankhah, A.M., Safaeian, A.R., et al., 2012. Effect of treatment with omega-3 fatty acids on C-reactive protein and tumor necrosis factor-alfa in hemodialysis patients. Saudi J. Kidney Dis. Transpl. 23 (3), 500–506.
Turkheimer, F.E., Veronese, M., Mondelli, V., Cash, D., Pariante, C.M., 2023. Sickness behaviour and depression: an updated model of peripheral-central immunity interactions. Brain Behav. Immun. 111, 202–210.
Uher, R., Tansey, K.E., Dew, T., Maier, W., Mors, O., Hauser, J., et al., 2014. An inflammatory biomarker as a differential predictor of outcome of depression treatment with escitalopram and nortriptyline. Am. J. Psychiatry 171 (12), 1278–1286.
White, J., Kivimäki, M., Jokela, M., Batty, G.D., 2017. Association of inflammation with specific symptoms of depression in a general population of older people: the english longitudinal study of ageing. Brain Behav. Immun. 61, 27–30.
Wijaya, M.T., Jin, R.R., Liu, X., Zhang, R., Lee, T.M.C., 2022. Towards a multidimensional model of inflamed depression. Brain Behav Immun Health. 26, 100564.
Wium-Andersen, M.K., Orsted, D.D., Nielsen, S.F., Nordestgaard, B.G., 2013. Elevated C-reactive protein levels, psychological distress, and depression in 73, 131 individuals. JAMA Psychiat. 70 (2), 176–184.
Wu, S.K., Chen, W.J., Chang, J.P., Guu, T.W., Hsin, M.C., Huang, C.K., et al., 2023. Personalized medicine of Omega-3 fatty acids in depression treatment in obese and metabolically dysregulated patients. J Pers Med. 13 (6).
Young, J.J., Silber, T., Bruno, D., Galatzer-Levy, I.R., Pomara, N., Marmar, C.R., 2016. Is there Progress? an overview of selecting biomarker candidates for major depressive disorder. Front. Psych. 7, 72.
Zhou, L., Xiong, J.Y., Chai, Y.Q., Huang, L., Tang, Z.Y., Zhang, X.F., et al., 2022. Possible antidepressant mechanisms of omega-3 polyunsaturated fatty acids acting on the central nervous system. Front. Psych. 13, 933704.